

Mathematical Statistics

Lecture 2: Introduction to Probability Counting Coin Flips and Dice

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1 The Fundamental Definition of Probability

1.1 What is Probability?

At its core, probability is a mathematical framework for quantifying how likely an event is to happen. It gives us a number between 0 (impossible) and 1 (certain) to describe the chance of an outcome.

🎯 Key Idea: The Definition of Probability

The probability of an event, let's call it event **A**, is calculated by a simple ratio. It is the number of ways that event **A** can occur, divided by the total number of possible outcomes.

The Formula:

$$P(A) = \frac{\text{Number of ways A can happen}}{\text{Total number of possible outcomes}}$$

This simple formula is the foundation for everything we will do. At its heart, probability is just a very clever and systematic way of **counting**.

1.2 Important Assumptions

For these introductory examples, we make a few simplifying assumptions that are crucial to state:

- **Fairness:** The coin is "fair," meaning heads and tails have an equal chance of occurring. The dice are "fair," meaning each of the six faces has an equal chance of landing up.
- **Independence:** Each event is independent of the others. The result of the first coin flip has no effect on the second. The roll of the first die does not influence the roll of the second.

We will relax these assumptions later in the course to tackle more complex, real-world problems.

2 Worked Example 1: Flipping a Coin Twice

Problem: I flip a fair coin twice.

1. What is the probability of getting **at least one heads**? (Event A)
2. What is the probability of getting **no heads**? (Event B)

2.1 Step-by-Step Solution

Step 1: Determine the Total Number of Possible Outcomes

First, we list every single possible outcome of flipping a coin twice. This complete set of outcomes is called the **Sample Space**, often denoted by Ω . Let H = Heads and T = Tails.

- The possible outcomes are: (Heads, Heads), (Heads, Tails), (Tails, Heads), (Tails, Tails).
- Our Sample Space is $\Omega = \{HH, HT, TH, TT\}$.
- The **total number of possible outcomes is 4**.

Step 2: Calculate the Probability of Event A (At least one heads)

Now we count how many outcomes satisfy our condition.

- The set for event A is $A = \{HH, HT, TH\}$.
- The **number of ways A can happen is 3**.

Now, we apply the formula:

$$P(A) = \frac{\text{Number of ways for A}}{\text{Total Outcomes}} = \frac{3}{4} = 0.75 \text{ or } 75\%$$

Step 3: Calculate the Probability of Event B (No heads)

We repeat the process for the second question.

- The set for event B is $B = \{TT\}$.
- The **number of ways B can happen is 1**.

Applying the formula:

$$P(B) = \frac{\text{Number of ways for B}}{\text{Total Outcomes}} = \frac{1}{4} = 0.25 \text{ or } 25\%$$

3 Worked Example 2: Rolling Two Dice

Problem: I roll two fair, six-sided dice. What is the probability that **at least one of the dice is a 5?** (Event A)

3.1 Step 1: Determine the Total Number of Possible Outcomes

- The first die can land on any of 6 numbers.
- For *each* of those outcomes, the second die can also land on any of 6 numbers.
- Therefore, the total number of unique combinations is $6 \times 6 = 36$.

3.2 Step 2: Calculate the Probability of Event A (At least one 5)

3.2.1 Method 1: Direct Counting (Brute Force)

Let's list every outcome where at least one die is a 5:

- **If the first die is NOT a 5:** (1, 5), (2, 5), (3, 5), (4, 5), (6, 5) — *5 outcomes*
- **If the first die IS a 5:** (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6) — *6 outcomes*

The total number of ways A can happen is $5 + 6 = 11$. Applying the formula:

$$P(A) = \frac{\text{Number of ways for A}}{\text{Total Outcomes}} = \frac{11}{36}$$

3.2.2 Method 2: Logical Reasoning

We can satisfy the condition "at least one 5" in two distinct, non-overlapping ways:

1. The first die is a 5.
2. The first die is NOT a 5, **but** the second die IS a 5.

The total probability is the sum of the probabilities of these two cases:

$$P(A) = P(\text{Die 1 is 5}) + P(\text{Die 1 is NOT 5}) \times P(\text{Die 2 is 5})$$

$$P(A) = \frac{1}{6} + \left(\frac{5}{6} \times \frac{1}{6}\right) = \frac{1}{6} + \frac{5}{36}$$

To add these fractions, we find a common denominator:

$$P(A) = \frac{6}{36} + \frac{5}{36} = \frac{11}{36}$$

4 Big Picture Concepts

4.1 Deterministic Systems vs. Random Models

☉ Key Idea: What Do We Mean By "Random"?

Is a coin flip truly random? In a classical physics sense, no. It is a deterministic system governed by physics. We call it "random" because the outcome is extremely sensitive to initial conditions that are **too complex for us to measure or model perfectly**. Probability is our mathematical tool for modeling this perceived randomness.

Modeled as Random: Most things we study (weather, turbulence, economics, coin flips) are deterministic but chaotic systems that we model effectively using probability.

4.2 The Forward and Inverse Problem: Probability vs. Statistics

Probability vs. Statistics

- **Probability (The Forward Problem):** We assume a model is true (e.g., "This coin is fair") and then calculate the likelihood of future events.
Question: Given a model, what will the data look like?
- **Statistics (The Inverse Problem):** We observe data first (e.g., "I got 10 heads in a row") and then use that data to make inferences about the underlying model.

Question: Given the data, what is the best model?

5 Summary of Lecture 2

≡ Lecture Summary

1. **Probability** is fundamentally about counting: comparing the number of ways a specific event can happen to the total number of possible outcomes.
2. The key formula is $P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$.
3. We solved two classic problems: the probability of at least one heads in two coin flips (**3/4**) and the probability of at least one 5 in two dice rolls (**11/36**).
4. Many systems we model as **random** are actually complex deterministic systems. We use probability because we lack the information to make perfect predictions.
5. **Probability** predicts data from a model, while **Statistics** infers a model from data.